Web Design and Developmentlevel-3(Batch-6)
BCTI, KHULNA

1. Describe the usage of mail merge in word processing software.
Mail Merge is used to create personalized letters or documents for multiple recipients by linking a main document to a data source (like Excel).

2. Difference between <section>, <article>, and .
- **<section>:** Represents a thematic grouping of content.
- **<article>:** Represents a self-contained, independent piece of content (e.g., a blog post).
- ** :** A generic container used for styling purposes without semantic meaning.

3. Difference between raster and vector image.
- **Raster**: Made of **pixels**. They lose quality when resized and zoomed (e.g., JPEG, PNG).
- **Vector**: Made of **mathematical paths** and remain sharp at any size. (e.g., SVG, AI).

4. Difference between relative and absolute units in CSS.
- **Relative units**: Relative units (%, em, rem) change based on the screen or parent size.
- **Absolute units**: Absolute units (px, cm) are fixed and do not change.

5. Describe array and forEach() method in JavaScript.
- **Array**: An **Array** is a collection of data stored in a single variable.
- **forEach()**: The **forEach()** method is used to execute a function once for each element in that array.

6. Describe addEventListener vs onclick handler.
- **onclick**: **onclick** can only hold one event at a time and overwrites previous ones.
- **addEventListener**: **addEventListener** allows attaching multiple events to one element and offers more control.

7. Describe forEach() vs map().
- **forEach():** simply iterates over an array (returns nothing).
- **map():** iterates and returns a new array with modified values.

8. Explain Z-index property.
A CSS property that controls the stack order of elements. Elements with a higher z-index appear on top of those with a lower one.

9. Difference between == vs ===.
- == checks for value equality only (performs type coercion).
- === checks for both value and data type (strict equality).

10. Difference between CSS Framework vs. Custom CSS.
- **CSS Framework**: A Framework (Bootstrap, Tailwind) provides pre-written code for speed.
- **Custom CSS**: Custom CSS is written from scratch for unique, specific designs.

11. Difference among CSS position Absolute vs. Relative vs. Fixed.
- **Relative**: Positioned relative to its **normal position**.
- **Absolute**: Positioned relative to its **nearest positioned ancestor**.
- **Fixed**: Positioned relative to the **browser viewport**; it stays in place even when scrolling.

12. Purpose of the bind() method in JavaScript.
A JavaScript method used to create a new function that, when called, has its this keyword set to a specific value.

13. Use of media queries in responsive web design.
Media Queries: CSS rules used to apply different styles for different devices or screen sizes (Responsive Design)

14. Purpose of try...catch in JavaScript.
A block used to handle errors. The code in **try** is executed; if an error occurs, the code in **catch** handles it without crashing the program.

15. Synchronous vs. Asynchronous.
- **Synchronous**: Synchronous code runs line-by-line (blocking).
- **Asynchronous**: Asynchronous code (like API calls) allows other code to run while waiting for a task to finish.

16. Purpose of the DOM in JavaScript interaction.
DOM (Document Object Model): A programming interface that represents an HTML document as a tree structure, allowing JavaScript to manipulate the content and style.

17. Use of for...of loop with an example.
The for...of loop iterates over "iterable" objects like arrays or strings.
- *Example:* for (let x of [1, 2]) { console.log(x); }

18. How do VPNs enhance static and dynamic websites?
VPNs enhance security by encrypting the connection, protecting user data on both static and dynamic sites from hackers.

19. How does an interactive website differ from static and dynamic websites?
- **Static**: Fixed content.
- **Dynamic**: Content changes based on a database.
- **Interactive**: Responds instantly to user actions (like games or live chats).

20. Which code editor do you prefer and why?
VS Code (Visual Studio Code) is preferred due to its massive extension library, built-in terminal, and Git integration.

21. Can you explain the basic purpose of FTP software in web development?
FTP (File Transfer Protocol): Software used to upload website files from a local computer to a web server so the site can be live online.

22. How do you create an HTML login form with a submit button?
Created using the <form> tag containing <input type="text"> for username, <input type="password">, and <button type="submit">.

23. When should I consider using an SVG image?
Use SVG for logos, icons, and simple illustrations that need to be scaled without losing quality.

24. What is an ID selector in CSS?
A CSS selector that targets a single unique element using the # symbol (e.g., #header).

25. How do you apply multiple CSS classes to a single HTML element?
Add them to the class attribute separated by a space: .

26. How do you create a function in JavaScript?
Defined using the function keyword: function myFunc() { ... }.

27. Difference between function declaration and function expression?

- **Declaration:** A **Declaration** is a named function written as a statement.
- **Expression:** An **Expression** is a function assigned to a variable.

28. What are Array and Object in JavaScript?

- **Array:** An **Array** is a list of items [1, 2].
- **Object:** An **Object** is a collection of properties {key: value}.

29. Difference among for, while, and do-while loops?

- **for:** runs for a specific count.
- **while:** runs as long as a condition is true.
- **do-while:** runs at least once before checking the condition.

30. How do you handle conditional statements in JavaScript?
Handled using **if, else if, else,** and **switch** blocks.

31. How do you create a new Google Sheet in Google Drive?
Go to Google Drive, click **"+ New"**, and select **"Google Sheets."**

32. What is phpMyAdmin?
phpMyAdmin: phpMyAdmin is a free, open-source web-based tool used to manage **MySQL** and **MariaDB** databases.

33. Create function in js.With example.
JS Function Example: function greet(name) { return "Hello " + name; }.

34. What are Array and Object in JavaScript?

- **Array:** An **Array** is a list of items [1, 2].
- **Object:** An **Object** is a collection of properties {key: value}.

35. How do you handle conditional statements in JavaScript?
Handled using **if, else if, else,** and **switch** blocks.

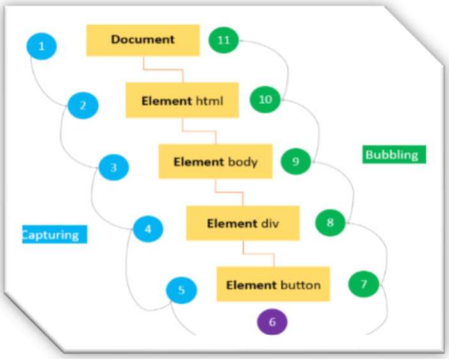
36. What is the purpose of a loop in JS?
To automate repetitive tasks by running a block of code multiple times.

37. What is the Document Object Model (DOM) in JS?
DOM (Document Object Model): A programming interface that represents an HTML document as a tree structure, allowing JavaScript to manipulate the content and style.

38. Explain the concept of strict mode in JS.
Strict Mode: Enabled by writing **"use strict";**. It catches common coding mistakes and prevents the use of unsafe syntax.

39. What is the concept of event bubbling and event capturing in JS?
Event Bubbling and Capturing are the two ways an event propagates through the DOM tree:

- **Event Capturing:** The event starts from the **top-most parent** (Window/Document) and moves **downwards** to the target element.
- **Event Bubbling:** The event starts from the **target element** and moves **upwards** through its parents to the top (Default behavior in JS).
- **Note:** By default, most event handlers use bubbling.
- ***Example:***



40. What role do search engines play in indexing and retrieving information from the internet?
Search Engines: They crawl and index web pages to help users find relevant information quickly through search queries.

41. How does a static website differ from a dynamic website?
Static vs. Dynamic Difference: Static sites look the same for everyone; **Dynamic** sites show personalized data (like a user profile).

42. What are some popular frameworks and platforms used for developing dynamic websites?
Popular Frameworks: React, Angular, Vue (Frontend); Node.js, Laravel, Django (Backend/Dynamic).

43. What are the main features and functionality of an interactive website?
Interactive Website Features: Real-time feedback, user accounts, search bars, and animations that react to clicks/hovers.

44. How do you ensure your design is compatible with both desktop and mobile browsers?
Device Compatibility: Use **Responsive Design** (Flexible grids and Media Queries) and test across different browsers.

45. Write down the name of some key features you look for in a code editor for web design development task.
Code Editor Features: Syntax highlighting, Autocomplete (IntelliSense), Extension support, and Debugging tools.

46. What are RESTful APIs?
RESTful APIs: A set of rules that allow different software systems to communicate over the web using HTTP methods (GET, POST, etc.).

47. What is phpMyAdmin, and how is it used in web development?
phpMyAdmin: phpMyAdmin is a free, open-source web-based tool used to manage **MySQL** and **MariaDB** databases.

48. Difference between Synchronous and Asynchronous

Synchronous: Synchronous code waits for each job to complete. (সিনক্রোনাস কোড লাইনের পর লাইন ধারাবাহিকভাবে চলে। আগের কাজ শেষ হওয়া পর্যন্ত অপেক্ষা করে।)

Asynchronous: Asynchronous code continues execution while waiting for the background task to finish.(অ্যাসিনক্রোনাস কোড কোনো কাজের জন্য অপেক্ষা না করে পরবর্তী কোডগুলো চালিয়ে দেয়।)

49. What is AJAX in JavaScript?

AJAX: AJAX (Asynchronous JavaScript and XML) is a technique used in web development to update parts of a web page without reloading the entire page. (AJAX হলো ওয়েব ডেভেলপমেন্টের একটি কৌশল, যার মাধ্যমে পুরো পেজটি রিলোড (Reload) না করেই ওয়েবসাইটের নির্দিষ্ট কিছু অংশ আপডেট বা পরিবর্তন করা যায়।)

50. What is the Document Object Model (DOM) in JS?

DOM: DOM (Document Object Model) A programming interface that represents an HTML document as a tree structure, allowing JavaScript to manipulate the content and style. (DOM হলো একটি প্রোগ্রামিং ইন্টারফেস যা একটি HTML ডকুমেন্টকে একটি ট্রি (Tree) কাঠামোর মতো উপস্থাপন করে। এর ফলে জাভাস্ক্রিপ্ট ব্যবহার করে খুব সহজেই ওয়েবসাইটের কন্টেন্ট (লেখা, ছবি) এবং স্টাইল (ডিজাইন) পরিবর্তন বা নিয়ন্ত্রণ করা যায়।)

51. What is BOM in JavaScript?

BOM: BOM (Browser Object Model) is a set of objects provided by the browser that allow JavaScript to interact with the browser window itself, rather than just the content of the web page. (BOM হলো ব্রাউজারের এমন কিছু অবজেক্টের সমষ্টি যার মাধ্যমে জাভাস্ক্রিপ্ট সরাসরি ব্রাউজারের সাথে যোগাযোগ করতে পারে। এটি HTML ডকুমেন্টের বাইরে ব্রাউজারের উইন্ডো, ইউআরএল (URL), হিস্ট্রি এবং স্ক্রিন সাইজ নিয়ে কাজ করে।)

52. What is Manipulate in JavaScript?

Manipulate: In JavaScript, **Manipulate** means to **change, add, or remove** elements on a webpage dynamically using code, without the user having to refresh the page. (জাভাস্ক্রিপ্টে 'Manipulate' (ম্যানিপুলেট) করা বলতে বোঝায়– পেজটি রিফ্রেশ না করেই কোড ব্যবহার করে ডাইনামিকভাবে ওয়েবপেজের বিভিন্ন উপাদান (elements) পরিবর্তন করা, নতুন কিছু যোগ করা অথবা কোনো কিছু মুছে ফেলা।)

53. What is Condition in JavaScript

Condition: In JavaScript, a **condition** is a logical expression that evaluates to either **true** or **false**. (জাভাস্ক্রিপ্টে কন্ডিশন (Condition) বা শর্ত হলো একটি লজিক্যাল এক্সপ্রেশন (logical expression), যার ফলাফল শুধুমাত্র সত্য (true) অথবা মিথ্যা (false) হতে পারে।)